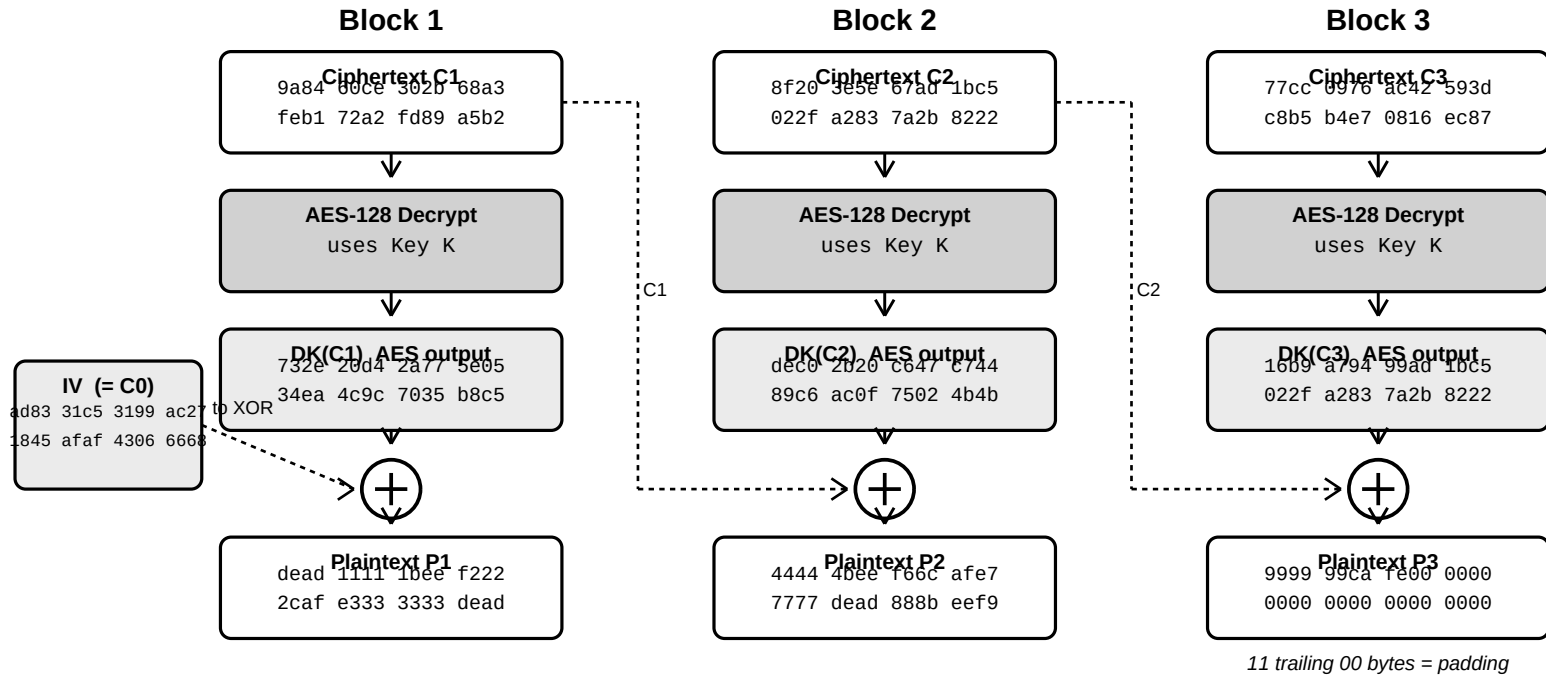


Part A — AES-128-CBC Decryption

Formula: $P_i = DK(C_i) \text{ XOR } C_{(i-1)}$ with $C_0 = IV$

Key K = 26671722 51915cc4 4cfc2ec3 a88b4d89

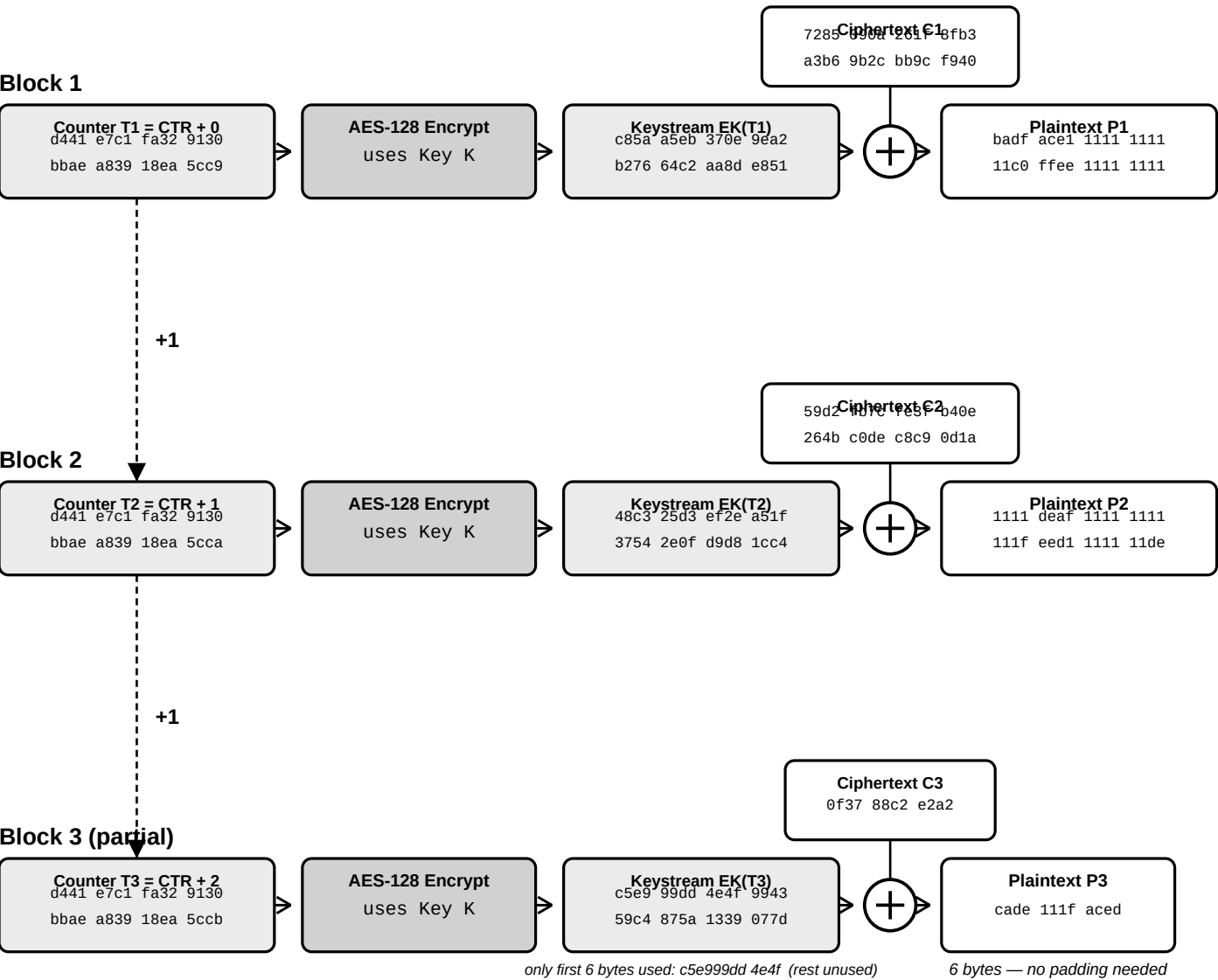


All values in hexadecimal. Diagram is original (monochrome). Chaining uses prior ciphertext (or IV), not prior plaintext.

Part B — AES-128-CTR Decryption

Formulas: $T_i = \text{CTR} + (i - 1)$ $P_i = C_i \text{ XOR } EK(T_i)$

Key K = 53153913 a1623f8a 089ee173 60fb9ac9



All values in hexadecimal. Counter increments by 1 each block. Block 3 ciphertext is only 6 bytes.

CS3002 — Assignment 2: Block Cipher Modes Decryption

Recovered plaintexts and discussion answers

Recovered Plaintext — Message 1 (AES-128-CBC)

```
dead 1111 1bee f222 2caf e333 3333 dead
4444 4bee f66c afe7 7777 dead 888b eef9
9999 99ca fe00 0000 0000 0000 0000 0000
```

Full (48 bytes): dead11111beef2222cafe33333333dead44444beef66cafe77777dead888beef9999999cafe000000000000000000000000
Last 11 bytes are 00 padding so the final CBC block is a full 16 bytes.

Question 4 — Can CBC decryption be parallelized?

Yes. CBC encryption must run in order because each new ciphertext block needs the previous ciphertext. CBC decryption is different. All ciphertext blocks C1, C2, C3 are already known as inputs. Each AES decrypt step $DK(C_i)$ only needs that block C_i and the shared key, so those AES operations can run at the same time. After that, XOR with IV or $C_{(i-1)}$ also uses only known ciphertext. The CBC diagram shows each AES decrypt box taking only its own C_i — it does not wait for the previous plaintext. So the sequential limit of CBC encryption does not apply to decryption in the same way.

Recovered Plaintext — Message 2 (AES-128-CTR)

```
badf ace1 1111 1111 11c0 ffee 1111 1111
1111 deaf 1111 1111 111f eed1 1111 11de
cade 111f aced
```

Full CTR plaintext (hex): badface111111111111c0ffee111111111111deaf111111111111feed11111111decade111faced
Length: 38 bytes. No padding bytes.

Question 6 — Why is padding not required in CTR mode?

CTR mode turns AES into a stream cipher. It encrypts a counter to make a keystream, then XORs that keystream with the data. The ciphertext can be the same length as the plaintext. If the last piece is shorter than 16 bytes, only that many keystream bytes are used. In this message the last ciphertext is 0f3788c2e2a2 (6 bytes), so only the first 6 keystream bytes are XORed and the rest are unused. CBC works on full 16-byte blocks, so leftover space must be filled with padding (here trailing zero bytes on the CBC plaintext). CTR does not need that.

Method: AES-128 block encrypt/decrypt with CBC chaining and CTR counter increment, plus XOR.